

# Rockchip RK3566\_RK3568 Linux SDK Quick Start

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## Preface

### Overview

The document presents Rockchip RK3566/RK3568 Linux SDK release notes, aiming to help engineers get started with RK3566/RK3568 Linux SDK development and debugging faster.

### Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

### Chip System Support Status

| Chip Name      | Uboot Version | Kernel Version | Debian Version | Buildroot Version |
|----------------|---------------|----------------|----------------|-------------------|
| RK3566, RK3568 | 2017.9        | 5.10, 6.1      | 11, 12         | 2021.11, 2024.02  |

### Revision Record

| Date       | Version | Author      | Modification Description           |
|------------|---------|-------------|------------------------------------|
| 2022-06-20 | V1.0.0  | Caesar Wang | Initial version.                   |
| 2022-09-20 | V1.0.1  | Caesar Wang | Added support for Linux 5.10.      |
| 2022-11-20 | V1.0.2  | Caesar Wang | Updated flashing instructions.     |
| 2023-04-20 | V1.1.0  | Caesar Wang | Adapted to the new version of SDK. |
| 2023-05-20 | V1.1.1  | Caesar Wang | Updated SDK to V1.1.1.             |
| 2023-06-20 | V1.2.0  | Caesar Wang | Updated SDK to V1.2.0.             |
| 2023-09-20 | V1.3.0  | Caesar Wang | Updated SDK to V1.3.0.             |
| 2023-12-20 | V1.4.0  | Caesar Wang | Updated SDK to V1.4.0.             |
| 2024-06-20 | V1.5.0  | Caesar Wang | Updated SDK to V1.5.0.             |
| 2025-12-20 | V1.6.0  | Caesar Wang | Updated SDK to V1.6.0.             |

## Contents

### Rockchip RK3566\_RK3568 Linux SDK Quick Start

1. Precompiled SDK Images
2. Development Environment Setup
  - 2.1 Preparing the Development Environment
  - 2.2 Installing Libraries and Toolkits
    - 2.2.1 Checking and Upgrading the Host's Python Version
    - 2.2.2 Checking and Upgrading the Host's `make` Version
    - 2.2.3 Checking and Upgrading the Host's LZ4 Version
3. Docker Environment Setup
4. Software Development Guide
  - 4.1 Development Guide
  - 4.2 Chip Documentation
  - 4.3 Buildroot Development Guide
  - 4.4 Debian Development Guide
  - 4.5 Third-Party OS Porting
  - 4.6 RKNPU Development Guide
    - 4.6.1 RKNN-TOOLKIT2
    - 4.6.2 RKNN Runtime
    - 4.6.3 RKNN Driver
  - 4.7 DPDK Development Guide
  - 4.8 Real-time Linux Development Guide
    - 4.8.1 Pressure Testing
      - 4.8.1.1 PREEMPT\_RT Patch
      - 4.8.1.2 Xenomai Cobalt Mode
  - 4.9 Software Update Log
5. Hardware Development Guide
6. IO Power Supply Design Considerations
7. SDK Configuration Framework Description
  - 7.1 SDK Project Directory Overview
8. Introduction to Cross-Compilation Toolchain for SDK
  - 8.1 U-Boot and Kernel Compilation Toolchain
  - 8.2 Buildroot Toolchain
    - 8.2.1 Configuring the Compilation Environment
    - 8.2.2 Packaging Toolchain
  - 8.3 Debian Toolchain
  - 8.4 Yocto Toolchain
9. SDK Compilation Instructions
  - 9.1 Viewing SDK Compilation Commands
  - 9.2 SDK Board-Level Configuration
  - 9.3 Custom SDK Configuration
  - 9.4 Fully Automated Compilation
  - 9.5 Module Compilation
    - 9.5.1 U-Boot Compilation
    - 9.5.2 Kernel Compilation
    - 9.5.3 Recovery Compilation
    - 9.5.4 Buildroot Compilation
      - 9.5.4.1 Building Buildroot Modules
    - 9.5.5 Debian Compilation
    - 9.5.6 Yocto Compilation
  - 9.6 Firmware Packaging

## 10. Flashing Instructions

### 10.1 Windows Flashing Instructions

### 10.2 Linux Flashing Instructions

### 10.3 System Partition Description

# 1. Precompiled SDK Images

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Developers can bypass the process of compiling the entire operating system from source code by using the precompiled RK3566\_RK3568 Linux SDK image. This allows for a quick start to development and related assessments and comparisons, reducing the time and cost wasted due to compilation issues.

The SDK firmware can be downloaded from the public address [SDK Firmware Download Here](#).

For Linux 5.10 firmware, the path is: **General Linux SDK Firmware -> Linux 5.10 -> RK3566\_RK3568**

For Linux 6.1 firmware, the path is: **General Linux SDK Firmware -> Linux 6.1 -> RK3566\_RK3568**

If modifications to the SDK code or a quick start are needed, please refer to the following sections.

## 2. Development Environment Setup

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### 2.1 Preparing the Development Environment

We recommend using a system with Ubuntu 22.04 or a higher version for compilation. Other Linux versions may require corresponding adjustments to the software packages. In addition to system requirements, there are other hardware and software requirements.

Hardware Requirements: 64-bit system with more than 40GB of hard disk space. If you are performing multiple builds, you will need even more disk space.

Software Requirements: A system with Ubuntu 22.04 or a higher version.

### 2.2 Installing Libraries and Toolkits

When developing device applications via the command line, you can install the libraries and tools required for compiling the SDK through the following steps.

Use the following apt-get command to install the libraries and tools needed for subsequent operations:

```
sudo apt-get update && sudo apt-get install git ssh make gcc libssl-dev liblz4-tool expect expect-dev g++ patchelf chrpath gawk texinfo chrpath diffstat binfmt-support qemu-user-static live-build bison flex fakeroot cmake gcc-multilib g++-multilib unzip device-tree-compiler ncurses-dev libgucharmap-2-90-dev bzip2 expat gpgv2 cpp-aarch64-linux-gnu libgmp-dev libmpc-dev bc python-is-python3 python3-setuptools swig python3-dev
```

#### Note:

The installation command is applicable for Ubuntu 22.04. For other versions, please use the corresponding installation commands based on the package names. If you encounter errors during compilation, you can install the corresponding software packages based on the error messages. Among

them:

- Python requires the installation of version 3.6 or higher, with version 3.6 used as an example here.
- Make requires the installation of version 4.0 or higher, with version 4.2 used as an example here.
- lz4 requires the installation of version 1.7.3 or higher.
- Compiling Yocto requires a VPN network.

## 2.2.1 Checking and Upgrading the Host's Python Version

The method for checking and upgrading the host's Python version is as follows:

- Check the host's Python version

```
$ python3 --version
Python 3.10.6
```

If the requirement of Python version  $\geq 3.6$  is not met, you can upgrade by the following method:

- Upgrade to the new version of Python 3.6.15

```
PYTHON3_VER=3.6.15
echo "wget https://www.python.org/ftp/python/${PYTHON3_VER}/Python-${PYTHON3_VER}.tgz"
echo "tar xf Python-${PYTHON3_VER}.tgz"
echo "cd Python-${PYTHON3_VER}"
echo "sudo apt-get install libsqlite3-dev"
echo "./configure --enable-optimizations"
echo "sudo make install -j8"
```

## 2.2.2 Checking and Upgrading the Host's make Version

The method for checking and upgrading the host's make version is as follows:

- Checking the host's make version

```
$ make -v
GNU Make 4.2
Built for x86_64-pc-linux-gnu
```

- Upgrading to a newer version of make 4.2

```
$ sudo apt update && sudo apt install -y autoconf autopoint

git clone https://gitee.com/mirrors/make.git
cd make
git checkout 4.2
git am $BUILDR00T_DIR/package/make/*.patch
autoreconf -f -i
./configure
make make -j8
sudo install -m 0755 make /usr/bin/make
```

## 2.2.3 Checking and Upgrading the Host's LZ4 Version

The method to check and upgrade the host's LZ4 version is as follows:

- Check the host's LZ4 version

```
$ lz4 -v
*** LZ4 command line interface 64-bits v1.9.3, by Yann Collet ***
```

- Upgrade to a newer version of LZ4

```
git clone https://gitee.com/mirrors/LZ4_old1.git
cd LZ4_old1

make
sudo make install
sudo install -m 0755 lz4 /usr/bin/lz4
```

## 3. Docker Environment Setup

To assist developers in quickly completing the complex preparation work for the development environment mentioned above, we also provide a cross-compiler Docker image to enable customers to quickly verify, thereby reducing the time required to build the compilation environment.

Before using the Docker environment, you can refer to the following document for operations:

[https://<SDK>/docs/en/Linux/Docker/Rockchip\\_Developer\\_Guide\\_Linux\\_Docker\\_Deploy\\_EN.pdf](https://<SDK>/docs/en/Linux/Docker/Rockchip_Developer_Guide_Linux_Docker_Deploy_EN.pdf).

The following systems have been verified:

| Distribution Version | Docker Version | Image Load | Firmware Compilation |
|----------------------|----------------|------------|----------------------|
| Ubuntu 22.04         | 24.0.5         | pass       | pass                 |
| Ubuntu 21.10         | 20.10.12       | pass       | pass                 |
| Ubuntu 21.04         | 20.10.7        | pass       | pass                 |
| Ubuntu 18.04         | 20.10.7        | pass       | pass                 |
| Fedora 35            | 20.10.12       | pass       | NR (not run)         |

The Docker image can be obtained from the website [Docker Image](#).

## 4. Software Development Guide

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### 4.1 Development Guide

To assist development engineers in quickly becoming proficient with the SDK development and debugging process, the "Rockchip\_Developer\_Guide\_Linux\_Software\_EN.pdf" is released along with the SDK. It can be obtained in the `docs/en/RK3566_RK3568` directory and will be continuously improved and updated.

### 4.2 Chip Documentation

To assist development engineers in quickly becoming proficient with the development and debugging of RK3566 and RK3568, the chip manuals titled "Rockchip\_RK3566\_Datasheet\_V1.4-20240621.pdf" and "Rockchip\_RK3568\_Datasheet\_V2.1-20240621.pdf" are released alongside the SDK. They can be accessed in the `docs/en/RK3566_RK3568/Datasheet` directory and will be continuously improved and updated.

### 4.3 Buildroot Development Guide

To assist development engineers in quickly getting started with the development and debugging of the Buildroot system, the SDK release includes the "Rockchip\_Developer\_Guide\_Buildroot\_EN.pdf" development guide, which can be obtained in the `docs/en/Linux/System` directory and will be continuously improved and updated.

### 4.4 Debian Development Guide

To assist development engineers in quickly becoming proficient with the development and debugging of RK3566 and RK3568 Debian, the "Rockchip\_Developer\_Guide\_Debian\_EN.pdf" development guide is released with the SDK. It can be obtained under `docs/en/Linux/System` and will be continuously improved and updated.

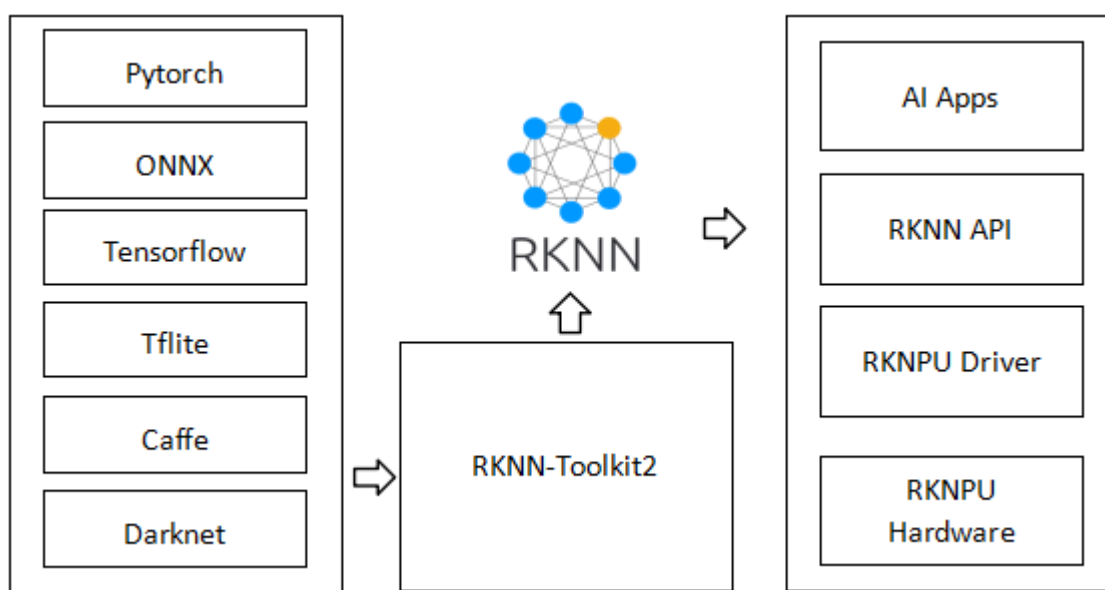


## 4.5 Third-Party OS Porting

To assist development engineers in quickly mastering and familiarizing themselves with the porting and adaptation of third-party operating systems for RK3566 and RK3568, the SDK release includes the "Rockchip\_Developer\_Guide\_Third\_Party\_System\_Adaptation\_EN.pdf" development guide, which can be obtained under [docs/en/Linux/System](#) and will be continuously improved and updated.

## 4.6 RKNPU Development Guide

RKNN software stack can help users to quickly deploy AI models to Rockchip chips. The overall framework is as follows:



In order to use RKNPU, users need to first run the RKNN-Toolkit2 tool on the computer, convert the trained model into an RKNN format model, and then inference on the development board using the RKNN C API or Python API.

### 4.6.1 RKNN-TOOLKIT2

RKNN-Toolkit2 is a software development kit for users to perform model conversion, inference and performance evaluation on PC and Rockchip NPU platforms.

RKNN-Toolkit-Lite2 provides Python programming interfaces for Rockchip NPU platform to help users deploy RKNN models and accelerate the implementation of AI applications.

The development kit is located in the [external/rknn-toolkit2](#) directory, mainly used to implement a series of functions such as model conversion, optimization, quantization, inference, performance evaluation, and accuracy analysis.

Basic functions are as follows:

| Function            | Description                                                                                                                                                                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Model Conversion    | Supports floating-point models of Pytorch / TensorFlow / TFLite / ONNX / Caffe / Darknet<br>Supports quantization-aware models (QAT) of Pytorch / TensorFlow / TFLite<br>Supports dynamic input models (dynamicization/native dynamic)<br>Supports large models |
| Model Optimization  | Constant folding / OP correction / OP Fuse&Convert / Weight sparsification / Model pruning                                                                                                                                                                      |
| Model Quantization  | Supports quantization types: asymmetric i8/fp16<br>Supports Layer / Channel quantization methods; Normal / KL/ MMSE quantization algorithms<br>Supports mixed quantization to balance performance and accuracy                                                  |
| Model Inference     | Supports model inference through the simulator on the PC<br>Supports model inference on the NPU hardware platform (board-level inference)<br>Supports batch inference, supports multi-input models                                                              |
| Model Evaluation    | Supports performance and memory evaluation of the model on the NPU hardware platform                                                                                                                                                                            |
| Accuracy Analysis   | Supports quantization accuracy analysis function (simulator/NPU)                                                                                                                                                                                                |
| Additional Features | Supports version/device query functions, etc.                                                                                                                                                                                                                   |

## 4.6.2 RKNN Runtime

The development documentation for RKNN Runtime is located in the `external/rknpu2` directory of the project. It is used to infer models generated by RKNN-Toolkit2 and provides C/C++ programming interfaces for the Rockchip NPU platform, assisting users in deploying RKNN models and accelerating AI application implementation.

For detailed usage instructions, refer to the documentation in the current `doc/` directory:

```

└─ 01_Rockchip_RKNPU_Quick_Start_RKNN_SDK_V2.3.2_EN.pdf
└─ 02_Rockchip_RKNPU_User_Guide_RKNN_SDK_V2.3.2_EN.pdf
└─ 03_Rockchip_RKNPU_API_Reference_RKNN_Toolkit2_V2.3.2_EN.pdf
└─ 04_Rockchip_RKNPU_API_Reference_RKNNRT_V2.3.2_EN.pdf
└─ 05_RKNN_Compiler_Support_Operator_List_V2.3.2.pdf
└─ RKNNToolKit2_API_Difference_With_Toolkit1-2.3.2.md
└─ RKNNToolKit2_OP_Support-2.3.2.md
...

```

### 4.6.3 RKNN Driver

The RKNN kernel driver handles interactions with NPU hardware and is integrated into the Rockchip kernel code under `drivers/rknpu`.

## 4.7 DPDK Development Guide

To assist development engineers in quickly getting started with DPDK development and debugging, the SDK release includes the "Rockchip\_Developer\_Guide\_Linux\_DPDK\_EN.pdf" and "Rockchip\_Developer\_Guide\_Linux\_GMAC\_DPDK\_EN.pdf" development guides, which can be obtained in the `<SDK>/external/dpdk/rk_docs` directory and will be continuously improved and updated.

## 4.8 Real-time Linux Development Guide

As the demand for real-time performance in products increases, the real-time capabilities of standard Linux are no longer sufficient for many products. It is necessary to optimize standard Linux to enhance its real-time performance, such as using PREEMPT\_RT and the Xenomai real-time system.

Below are the latency test results conducted on RK3566\_RK3568 with Buildroot, based on both PREEMPT\_RT and Xenomai, as follows:

### 4.8.1 Pressure Testing

```
stress-ng -c 4 --io 2 --vm 1 --vm-bytes 4M --timeout 1000000s
```

#### 4.8.1.1 PREEMPT\_RT Patch

```
rk3568_t:/ # /data/cyclictest -c 0 -m -n -t 4 -p 99
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 9.36 9.67 9.84 8/1152 3678
```

|              |      |        |           |      |    |      |    |      |    |      |     |
|--------------|------|--------|-----------|------|----|------|----|------|----|------|-----|
| T: 0 ( 3482) | P:99 | I:1000 | C:5892687 | Min: | 9  | Act: | 20 | Avg: | 25 | Max: | 123 |
| T: 1 ( 3483) | P:99 | I:1500 | C:3928444 | Min: | 9  | Act: | 20 | Avg: | 25 | Max: | 116 |
| T: 2 ( 3484) | P:99 | I:2000 | C:2946323 | Min: | 9  | Act: | 13 | Avg: | 26 | Max: | 120 |
| T: 3 ( 3485) | P:99 | I:2500 | C:2357050 | Min: | 10 | Act: | 20 | Avg: | 26 | Max: | 126 |

(2-hour test)

#### 4.8.1.2 Xenomai Cobalt Mode

```
rk3568_t:/ # /data/cyclictest -c 0 -m -n -t 4 -p 99
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 7.65 7.80 7.75 8/1118 3998

T: 0 ( 3492) P:99 I:1000 C:21069118 Min:      2 Act:      4 Avg:      13 Max:      85
T: 1 ( 3493) P:99 I:1500 C:14046070 Min:      2 Act:     13 Avg:      15 Max:      76
T: 2 ( 3494) P:99 I:2000 C:10534550 Min:      2 Act:      6 Avg:      16 Max:      70
T: 3 ( 3495) P:99 I:2500 C:8427637 Min:      2 Act:     15 Avg:     15 Max:      96
(Testing for 5 hours)
```

For more details, please refer to the development patch package and instructions in [docs/Patches/Real-Time-Performance/](#).

## 4.9 Software Update Log

- Software release version upgrades can be viewed through the engineering XML. The specific method is as follows:

```
.repo/manifests$ realpath rk356x_linux5.10_release.xml
# For example: The printed version number is v1.5.0, and the update time is 20240620
# <SDK>/.repo/manifests/rk356x_linux/rk356x_linux5.10_release_v1.5.0_20240620.xml
```

- The content of software release version upgrades can be viewed through the engineering text, refer to the engineering directory:

```
<SDK>/.repo/manifests/rk356x_linux/RK3566_RK3568_Linux5.10_SDK_Note.md
or
<SDK>/docs/en/RK3566_RK3568/RK3566_RK3568_Linux5.10_SDK_Note.md
```

- Board-side version information can be obtained in the following way:

```
buildroot:/# cat /etc/os-release
NAME=Buildroot
VERSION=linux-5.10-gen-rkr8
ID=buildroot
VERSION_ID=2021.11
PRETTY_NAME="Buildroot 2021.11"
OS="buildroot"
BUILD_INFO="xxx  Thur Jun 20 23:12:04 CST 2024 - rockchip_rk3568"
KERNEL="5.10 - rockchip_linux_defconfig"
```

## 5. Hardware Development Guide

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For hardware-related development, please refer to the user manual in the project directory:

Rockchip RK3566 and RK3568 Hardware Design Guide:

```
<SDK>/docs/en/RK3566_RK3568/Hardware/Rockchip_RK3566_Hardware_Design_Guide_V1.1_20220206_EN.pdf  
<SDK>/docs/en/RK3566_RK3568/Hardware/Rockchip_RK3568_Hardware_Design_Guide_V1.2_20211107_EN.pdf
```

Rockchip RK3566 and RK3568 EVB Hardware User Guide:

```
<SDK>/docs/en/RK3566_RK3568/Hardware/Rockchip_RK3568_EVB_User_Guide_V1.2_EN.pdf  
<SDK>/docs/en/RK3566_RK3568/Hardware/Rockchip_RK3566_EVB2_User_Guide_V1.1_EN.pdf
```

## 6. IO Power Supply Design Considerations

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**The IO level of the controller power domain must be consistent with the IO level of the connected peripheral chip, and the voltage configuration of software must be consistent with the voltage of hardware to avoid GPIO damage.**



### Note:

*About matching of GPIO power domain and IO level:*

*PMUIO0\_VDD, PMUIO1\_VDD, VCCIO1\_VDD, VCCIO2\_VDD, VCCIO3\_VDD, VCCIO4\_VDD, VCCIO5\_VDD, VCCIO6\_VDD, VCCIO7\_VDD, voltage of these GPIO power domain must be consistent with the IO level voltage of the connected peripheral to avoid GPIO damage.*

*Also need to note that the voltage configuration of software should be consistent with the voltage of hardware: For example, if hardware IO level is connected to 1.8V, the voltage configuration of software should be configured to 1.8V accordingly; if hardware IO level should be connected to 3.3V, and the voltage configuration of software should also be configured to 3.3V to avoid GPIO damage.*

---

For more information, please refer to:

```
<SDK>/docs/en/RK3566_RK3568/Rockchip_RK356X_Introduction_IO_Power_Domains_Configuration.pdf  
<SDK>/docs/en/Common/IO-DOMAIN/Rockchip_Developer_Guide_Linux_IO_DOMAIN_EN.pdf
```

## 7. SDK Configuration Framework Description

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### 7.1 SDK Project Directory Overview

The SDK project directory includes directories such as buildroot, debian, rtos, app, kernel, u-boot, device, docs, and external. It uses a manifest to manage the repository and the repo tool to manage each directory or its corresponding git projects.

- app: Contains upper-layer application APPs, mainly some application demos.
- buildroot: Root file system based on Buildroot (2021 or 2024) development.
- debian: Root file system based on Debian bullseye (11 or 12) development.
- device/rockchip: Contains chip-level board configurations and scripts and files for compiling and packaging firmware.
- docs: Contains development guidance documents, platform support lists, tool usage documents, and Linux development guides.
- external: Contains third-party related repositories, including display, audio and video, camera, network, and security.
- kernel: Contains code for kernel development.
- hal: Contains bare-metal development libraries based on RK HAL hardware abstraction layer for AMPAK solutions.
- output: Contains firmware information, compilation information, XML, and host environment generated each time.
- prebuilts: Contains cross-compilation toolchains.
- rkbin: Contains Rockchip-related binaries and tools.
- rockdev: Contains compiled output firmware, actually a soft link to `output/firmware`.
- rtos: Root file system based on RT-Thread 4.1 development.
- tools: Contains commonly used tools for Linux and Windows operating systems.
- u-boot: Contains U-Boot code based on the next-dev version for development.
- uefi: Contains UEFI code based on the edk2 V2.7 version for development.
- yocto: Root file system based on Yocto 4.0 or 5.0 development.

## 8. Introduction to Cross-Compilation Toolchain for SDK

---

Considering that the Rockchip Linux SDK is currently compiled only in the Linux PC environment, we have only provided the cross-compilation toolchain for Linux. The prebuilt toolchain in the prebuilt directory is for use with U-Boot and Kernel. Specific Rootfs should be compiled using their respective toolchains or third-party toolchains.

## 8.1 U-Boot and Kernel Compilation Toolchain

The SDK prebuilts directory contains cross-compiled tools currently used for U-Boot and Kernel compilation, as follows:

| Directory                                                                          | Description                     |
|------------------------------------------------------------------------------------|---------------------------------|
| prebuilts/gcc/linux-x86/aarch64/gcc-arm-10.3-2021.07-x86_64-aarch64-none-linux-gnu | GCC ARM 10.3.1 64-bit toolchain |
| prebuilts/gcc/linux-x86/arm/gcc-arm-10.3-2021.07-x86_64-arm-none-linux-gnueabi     | GCC ARM 10.3.1 32-bit toolchain |

You can download the toolchain from the following address:

[Click here](#)

## 8.2 Buildroot Toolchain

### 8.2.1 Configuring the Compilation Environment

To compile individual modules or third-party applications, a cross-compilation environment must be configured. For instance, the cross-compilation tools for RK3568 are located in the `buildroot/output/rockchip_rk3568/host/usr` directory. It is necessary to set the `bin/` directory of the tools and the `aarch64-buildroot-linux-gnu/bin/` directory as environment variables. Execute the script for automatic configuration of environment variables in the top-level directory:

```
source buildroot/envsetup.sh rockchip_rk3568
```

Enter the command to check:

```
cd buildroot/output/rockchip_rk3568/host/usr/bin
./aarch64-linux-gcc --version
```

At this point, the following information will be printed:

```
aarch64-linux-gcc.br_real (Buildroot -gc307c95550) 12.3.0
```

### 8.2.2 Packaging Toolchain

Buildroot supports the packaging of the built-in toolchain into a compressed package for third-party applications to compile independently. For detailed information on how to package the toolchain, please refer to the Buildroot official documentation:

```
buildroot/docs/manual/using-buildroot-toolchain.txt
```

In the SDK, you can directly run the following command to generate the toolchain package:

```
./build.sh bmake: sdk
```

The generated toolchain package is located in the `buildroot/output/*/images/` directory, named `aarch64-buildroot-linux-gnu_sdk-buildroot.tar.gz`, for users who require it. After extraction, the path to `gcc` will be:

```
./aarch64-buildroot-linux-gnu_sdk-buildroot/bin/aarch64-buildroot-linux-gnu-gcc
```

## 8.3 Debian Toolchain

Utilizing Docker on the host machine, compile using `gcc` or `dpkg-buildpackage`.

## 8.4 Yocto Toolchain

Refer to the following:

[Building your own recipes from first principles](#)

[New Recipe](#)

# 9. SDK Compilation Instructions

The SDK can be configured and compiled for specific functionalities using `make` or `./build.sh` with target arguments.

Refer to the compilation instructions in `device/rockchip/common/README.md`.

## 9.1 Viewing SDK Compilation Commands

`make help`, for example:

```
$ make help
menuconfig          - interactive curses-based configurator
oldconfig           - resolve any unresolved symbols in .config
synconfig           - Same as oldconfig, but quietly, additionally update deps
olddefconfig        - Same as synconfig but sets new symbols to their default value
savedefconfig       - Save current config to RK_DEFCONFIG (minimal config)
...
```



The actual execution of make is `./build.sh`

That is, you can also run `./build.sh <target>` to compile related functions, and you can view the specific compilation commands through `./build.sh help`.

```
##### Rockchip Linux SDK #####

Manifest: rk356x_linux5.10_release_v1.5.0_20240620.xml

Log colors: message notice warning error fatal

Usage: build.sh [OPTIONS]
Available options:
chip[:<chip>[:<config>]]      choose chip
defconfig[:<config>]         choose defconfig
config                        modify SDK defconfig
...
updateimg                    build update image
otapackage                   build A/B OTA update image
all                           build images
release                       release images and build info
save                          alias of release
all-release                   build and release images
allsave                       alias of all-release
shell                         setup a shell for developing
cleanall                     cleanup
clean[:module[:module]]...   cleanup modules
post-rootfs <rootfs dir>      trigger post-rootfs hook scripts
help                          usage

Default option is 'all'.
```

## 9.2 SDK Board-Level Configuration

Navigate to the project directory `<SDK>/device/rockchip/rk3566_rk3568`:

| Board-Level Configuration                     | Description                                                                                      |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------|
| rockchip_rk3566_evb2_lp4x_v10_32bit_defconfig | Suitable for RK3566 EVB with LPDDR4 development board, running a 32-bit root file system         |
| rockchip_rk3566_evb2_lp4x_v10_defconfig       | Suitable for RK3566 EVB with LPDDR4 development board                                            |
| rockchip_rk3568_evb1_ddr4_v10_32bit_defconfig | Suitable for RK3568 EVB with LPDDR4 development board, running a 32-bit root file system         |
| rockchip_rk3568_evb1_ddr4_v10_defconfig       | Suitable for RK3568 EVB with DDR4 development board                                              |
| rockchip_rk3568_evb8_lp4_v10_32bit_defconfig  | Suitable for RK3568 EVB with LPDDR4/RK860X development board, running a 32-bit root file system  |
| rockchip_rk3568_evb8_lp4_v10_defconfig        | Suitable for RK3568 EVB8 with LPDDR4/RK860X development board                                    |
| rockchip_rk3568_pcie_ep_lp4x_v10_defconfig    | Suitable for RK3568 PCIe EP standard card development and verification                           |
| rockchip_defconfig                            | Default configuration, which will be symbolically linked to a specific board-level configuration |

#### Method 1

Add the board-level configuration file after `./build.sh`, for example:

Select the board-level configuration for **RK3566 EVB with LPDDR4 development board running a 32-bit root file system**:

```
./build.sh device/rockchip/rk3566_rk3568/rockchip_rk3566_evb2_lp4x_v10_32bit_defconfig
```

Select the board-level configuration for **RK3566 EVB with LPDDR4 development board**:

```
./build.sh device/rockchip/rk3566_rk3568/rockchip_rk3566_evb2_lp4x_v10_defconfig
```

Select the board-level configuration for **RK3568 EVB with DDR4 development board running a 32-bit root file system**:

```
./build.sh device/rockchip/rk3566_rk3568/rockchip_rk3568_evb1_ddr4_v10_32bit_defconfig
```

Select the board-level configuration for **RK3568 EVB with DDR4 development board**:

```
./build.sh device/rockchip/rk3566_rk3568/rockchip_rk3568_evb1_ddr4_v10_defconfig
```

Select the board-level configuration for **RK3568 EVB with LPDDR4/RK860X development board**:

```
./build.sh device/rockchip/rk3566_rk3568/rockchip_rk3568_evb8_lp4_v10_defconfig
```

Select the board-level configuration for **RK3568 EVB with LPDDR4/RK860X development board running a 32-bit root file system**:

```
./build.sh device/rockchip/rk3566_rk3568/rockchip_rk3568_evb8_lp4_v10_32bit_defconfig
```

Select the board-level configuration for **RK3568 PCIe EP standard card development and verification**:

```
./build.sh device/rockchip/rk3566_rk3568/rockchip_rk3568_pcie_ep_lp4x_v10_defconfig
```

## Method 2

```
rk3566_rk3568$ ./build.sh lunch
Pick a defconfig:

1. rockchip_defconfig
2. rockchip_rk3566_evb2_lp4x_v10_32bit_defconfig
3. rockchip_rk3566_evb2_lp4x_v10_defconfig
4. rockchip_rk3568_evb1_ddr4_v10_32bit_defconfig
5. rockchip_rk3568_evb1_ddr4_v10_defconfig
6. rockchip_rk3568_evb8_lp4_v10_32bit_defconfig
7. rockchip_rk3568_evb8_lp4_v10_defconfig
8. rockchip_rk3568_pcie_ep_lp4x_v10_defconfig
9. rockchip_rk3568_uvc_evb1_ddr4_v10_defconfig
Which would you like? [1]:
```

## 9.3 Custom SDK Configuration

The SDK can be configured through `make menuconfig`, and the main configurable components are as follows:

```
(rockchip_rk3568_evb1_ddr4_v10_defconfig) Name of defconfig to save
[*] Rootfs (Buildroot|Yocto|Debian) --->
[*] Loader (U-Boot) --->
[ ] AMPAK (Asymmetric Multi-Processing System)
[*] Kernel (Embedded in an Android-style boot image) --->
    Boot (Android-style boot image) --->
[*] Recovery (based on Buildroot) --->
    *** Security feature depends on buildroot rootfs ***
    Extra partitions (oem, userdata, etc.) --->
    Firmware (partition table, misc image, etc.) --->
[*] Update (Rockchip update image) --->
    Others configurations --->
```

- **Rootfs:** The term Rootfs here represents the "root file system," where you can choose different root file system configurations such as Buildroot, Yocto, Debian, etc.
- **Loader (U-Boot):** This is the configuration for the bootloader, typically U-Boot, which is used to initialize

the hardware and load the main operating system.

- AMPAK: A multi-core heterogeneous boot solution suitable for application scenarios requiring real-time performance.
- Kernel: Here, kernel options are configured to customize the Linux kernel to suit one's own hardware and application needs.
- Boot: Configuration for the boot partition support format is done here.
- Recovery (based on Buildroot): This is the configuration for the recovery environment based on Buildroot, used for system recovery and upgrades.
- PCBA test (based on Buildroot): This is the configuration for a PCBA (Printed Circuit Board Assembly) test environment based on Buildroot.
- Security: Security features are enabled, including Secureboot activation methods, Optee storage methods, and writing keys.
- Extra partitions: Used to configure additional partitions.
- Firmware: Firmware-related options are configured here.
- Update (Rockchip update image): Used to configure options for Rockchip's complete firmware.
- Others configurations: Additional configuration options.

The `make menuconfig` configuration interface provides a text-based user interface to select and configure various options.

After the configuration is complete, use the `make savedefconfig` command to save these settings, and the customized compilation will be based on these settings.

Through the above configuration, you can choose different Rootfs/Loader/Kernel configurations for various customized compilations, allowing for flexible selection and configuration of system components to meet specific needs.

## 9.4 Fully Automated Compilation

Navigate to the root directory of the project and execute the following commands to automatically complete all compilations:

```
./build.sh all # Compiles only module code (u-Boot, kernel, Rootfs, Recovery)
               # Requires further execution of `./build.sh ./mkfirmware.sh` for firmware
packaging

./build.sh     # Compiles module code (u-Boot, kernel, Rootfs, Recovery)
               # Packages into a complete update.img upgrade package
               # All compilation information is copied and generated in the out directory
```

The default is Buildroot, but a different rootfs can be specified by setting the environment variable `RK_ROOTFS_SYSTEM`. `RK_ROOTFS_SYSTEM` currently supports three systems: buildroot, debian, and yocto.

For example, to generate a debain system, the following commands can be used:

```
export RK_ROOTFS_SYSTEM=debian
./build.sh
or
RK_ROOTFS_SYSTEM=debian ./build.sh
```

Note:

It is recommended to clean up previous compilation artifacts with each SDK update by running  
`./build.sh cleanall`.

## 9.5 Module Compilation

### 9.5.1 U-Boot Compilation

```
./build.sh uboot
```

### 9.5.2 Kernel Compilation

- Method One

```
./build.sh kernel
```

- Method Two

```
cd kernel
export CROSS_COMPILE=../prebuilts/gcc/linux-x86/aarch64/gcc-arm-10.3-2021.07-x86_64-aarch64-
none-linux-gnu/bin/aarch64-none-linux-gnu-
make ARCH=arm64 rockchip_linux_defconfig

make ARCH=arm64 rk3566-evb2-lp4x-v10-linux.img -j16
or
make ARCH=arm64 rk3568-evb1-ddr4-v10-linux.img -j16
or
make ARCH=arm64 rk3568-evb8-lp4-v10-linux.img -j16
```

- Method Three

```
cd kernel
export CROSS_COMPILE=aarch64-linux-gnu-
make ARCH=arm64 rockchip_linux_defconfig

make ARCH=arm64 rk3566-evb2-lp4x-v10-linux.img -j16
or
make ARCH=arm64 rk3568-evb1-ddr4-v10-linux.img -j16
or
make ARCH=arm64 rk3568-evb8-lp4-v10-linux.img -j16
```

### 9.5.3 Recovery Compilation

Navigate to the root directory of the project and execute the following command to automatically complete the compilation and packaging of Recovery.

```
<SDK># ./build.sh recovery
```

After compilation, the recovery.img will be generated in the Buildroot directory `output/rockchip_rk3568_recovery/images`.

Note: The recovery.img includes the kernel, so every time the Kernel is modified, Recovery needs to be repackaged. The method to repackage Recovery is as follows:

```
<SDK># source buildroot/envsetup.sh
<SDK># cd buildroot
<SDK># make recovery-reconfigure
<SDK># cd -
<SDK># ./build.sh recovery
```

Note: Recovery is a non-essential feature, and some board-level configurations may not set it up.

### 9.5.4 Buildroot Compilation

Navigate to the root directory of the project and execute the following command to automatically complete the compilation and packaging of the Rootfs:

```
./build.sh rootfs
```

After compilation, different formats of images are generated in the Buildroot directory `output/rockchip_rk3568/images`, with the default format being rootfs.ext4.

For specific details, refer to the Buildroot development documentation:

```
<SDK>/docs/en/Linux/System/Rockchip_Developer_Guide_Buildroot_EN.pdf
```

#### 9.5.4.1 Building Buildroot Modules

Configuration for different chips and target functionalities can be set using `source buildroot/envsetup.sh`.

```
$ source buildroot/envsetup.sh
Top of tree: rk3566_rk3568

You're building on Linux
Lunch menu...pick a combo:

57. rockchip_rk3566
58. rockchip_rk3566_32
59. rockchip_rk3566_recovery
```

```
60. rockchip_rk3566_rk3568_base
61. rockchip_rk3566_rk3568_ramboot
62. rockchip_rk3568
63. rockchip_rk3568_32
64. rockchip_rk3568_recovery
```

...

Which would you like? [1]:

The default selection is 57, `rockchip_rk3566`. Then proceed to the RK3566's Buildroot directory to start compiling the relevant modules.

The `rockchip_rk3566_32` is used for compiling a 32-bit root file system, while `rockchip_rk3566_recovery` is for compiling the Recovery module.

For instance, to compile the rockchip-test module, the common compilation commands are as follows:

Navigate to the buildroot directory

```
<SDK># cd buildroot
```

- Delete and recompile the rockchip-test

```
buildroot# make rockchip-test-recreate
```

- Rebuild the rockchip-test

```
buildroot# make rockchip-test-rebuild
```

- Delete the rockchip-test

```
buildroot# make rockchip-test-dirclean
or
buildroot# rm -rf output/rockchip_rk3566/build/rockchip-test-master/
```

## 9.5.5 Debian Compilation

```
./build.sh debian
```

After compilation, the `linaro-rootfs.img` is generated in the `debian` directory.

**Note: It is necessary to install the relevant dependency packages in advance:**

```
sudo apt-get install binfmt-support qemu-user-static live-build
sudo dpkg -i ubuntu-build-service/packages/*
sudo apt-get install -f
```

For specific details, please refer to the Debian Development Documentation:

```
<SDK>/docs/en/Linux/System/Rockchip_Developer_Guide_Debian_EN.pdf
```

## 9.5.6 Yocto Compilation

Navigate to the root directory of the project and execute the following command to automatically complete the compilation and packaging of Rootfs:

```
./build.sh yocto
```

After compilation, a rootfs.img is generated in the `yocto` directory under `build/latest`.

The default username for login is `root`. For more information about Yocto, please refer to the [Rockchip Wiki](#).

FAQ:

- If you encounter the following issue during the compilation above:

Please use a locale setting which supports UTF-8 (such as `LANG=en_US.UTF-8`). Python can't change the filesystem locale after loading so we need a UTF-8 when Python starts or things won't work.

Solution:

```
locale-gen en_US.UTF-8
export LANG=en_US.UTF-8 LANGUAGE=en_US.en LC_ALL=en_US.UTF-8
```

Alternatively, refer to [setup-locale-python3](#).

- If you encounter git permission issues that cause compilation errors.

Due to the addition of the CVE-2022-39253 security detection patch in the newer version of git, if an older version of Yocto is used, the poky needs to include the following to be fixed:

```
commit ac3eb2418aa91e85c39560913c1ddfd2134555ba
Author: Robert Yang <liezhi.yang@windriver.com>
Date:   Fri Mar 24 00:09:02 2023 -0700

    bitbake: fetch/git: Fix local clone url to make it work with repo

    The "git clone /path/to/git/objects_symlink" couldn't work after the following
    change:

    https://github.com/git/git/commit/6f054f9fb3a501c35b55c65e547a244f14c38d56
```

Or you can also revert the git version on the PC to V2.38 or an earlier version, for example:

```
$ sudo apt update && sudo apt install -y libcurl4-gnutls-dev

git clone https://gitee.com/mirrors/git.git --depth 1 -b v2.38.0
cd git
make git -j8
make install
sudo install -m 0755 git /usr/bin/git
```



## 9.6 Firmware Packaging

After compiling the various parts such as Kernel, U-Boot, Recovery, and Rootfs, navigate to the root directory of the project and execute the following command to automatically complete the packaging of all firmware into the `output/firmware` directory:

Firmware Generation:

```
./build.sh firmware
```

## 10. Flashing Instructions

The interface distribution diagram for RK3566 EVB2 is as follows:



The interface distribution diagram for RK3568 EVB1 development board is as follows:



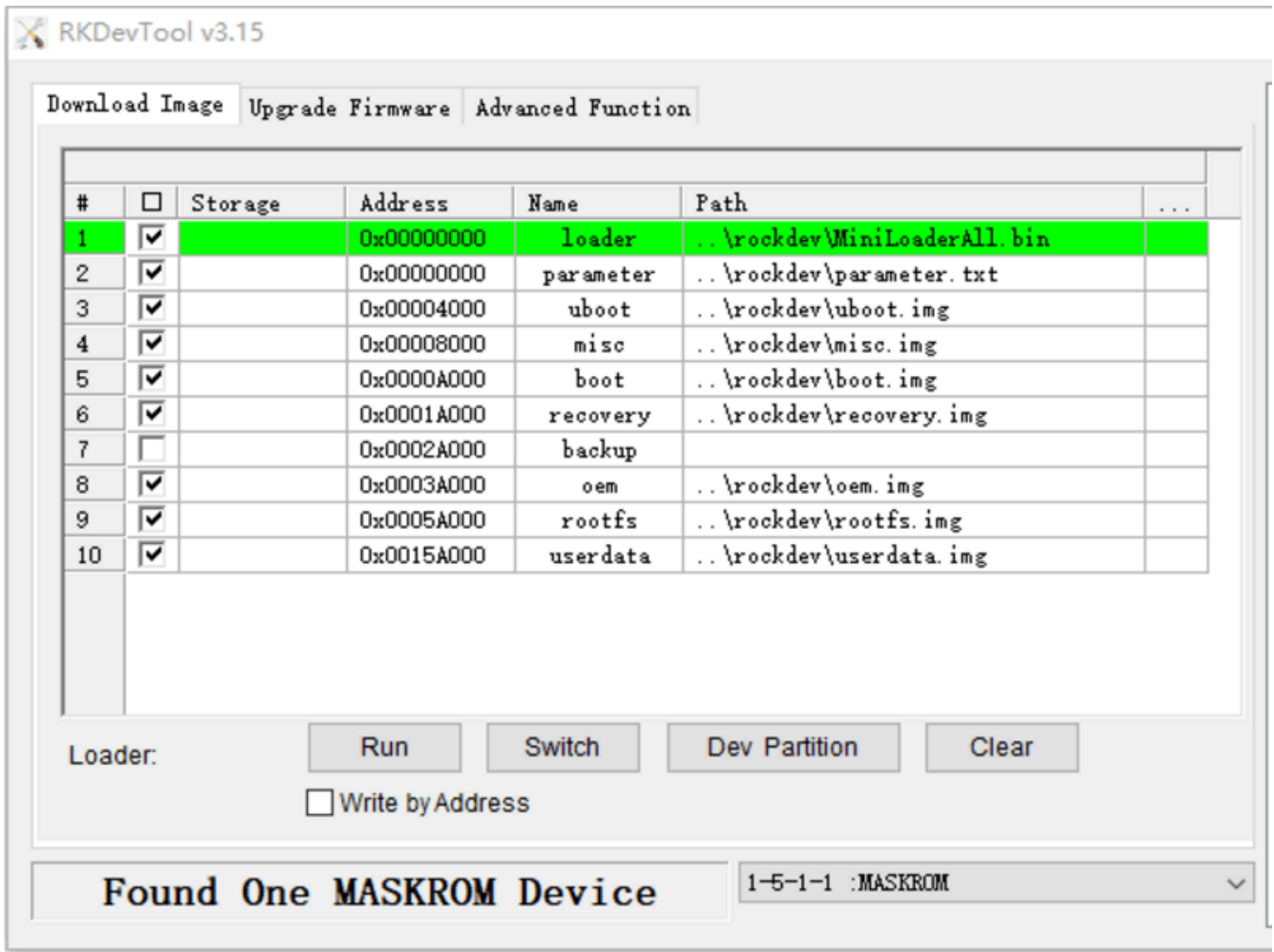
## 10.1 Windows Flashing Instructions

The SDK provides a Windows flashing tool (the tool version must be V3.31 or above), located in the root directory of the project:

```
tools/
├── windows/RKDevTool
```

As shown in the figure below, after compiling the corresponding firmware, the device must enter the MASKROM or BootROM flashing mode. Connect the USB download cable, hold the "MASKROM" button without releasing and press the reset button "RST", then release to enter the MASKROM mode. After loading the path of the compiled firmware, click "Execute" to perform the flashing. Alternatively, hold the "recovery" button without releasing and press the reset button "RST", then release to enter the loader mode for flashing. Below are the partition offsets for MASKROM mode and the flashing files.

(Note: The Windows PC needs to run the tool with administrator privileges to execute)



Note: Before flashing, the latest USB drivers must be installed. For more details on the drivers, see:

```
<SDK>/tools/windows/DriverAssitant_v5.13.zip
```

## 10.2 Linux Flashing Instructions

The flashing tool for Linux is located in the `tools/linux` directory (the `Linux_Upgrade_Tool` version must be V2.36 or above). Please ensure that your board is connected to MASKROM/loader rockusb. For example, if the firmware compiled is in the `rockdev` directory, the upgrade commands are as follows:

```
sudo ./upgrade_tool ul rockdev/MiniLoaderAll.bin -noreset
sudo ./upgrade_tool di -p rockdev/parameter.txt
sudo ./upgrade_tool di -u rockdev/uboot.img
sudo ./upgrade_tool di -misc rockdev/misc.img
sudo ./upgrade_tool di -b rockdev/boot.img
sudo ./upgrade_tool di -recovery rockdev/recovery.img
sudo ./upgrade_tool di -oem rockdev/oem.img
sudo ./upgrade_tool di -rootfs rockdev/rootfs.img
sudo ./upgrade_tool di -userdata rockdev/userdata.img
sudo ./upgrade_tool rd
```

Or upgrade the complete firmware after packaging:

```
sudo ./upgrade_tool uf rockdev/update.img
```

Or in the root directory, when the machine is running in MASKROM state, run the following upgrade:

```
./rkflash.sh
```

## 10.3 System Partition Description

Default Partition Description (Below is the RK3568 EVB partition reference)

| Number | Start (sector) | End (sector) | Size  | Name     |
|--------|----------------|--------------|-------|----------|
| 1      | 16384          | 24575        | 4M    | uboot    |
| 2      | 24576          | 32767        | 4M    | misc     |
| 3      | 32768          | 163839       | 64M   | boot     |
| 4      | 163840         | 425983       | 128M  | recovery |
| 5      | 425984         | 491519       | 32M   | backup   |
| 6      | 491520         | 13074431     | 6144M | rootfs   |
| 7      | 13074432       | 13336575     | 128M  | oem      |
| 8      | 13336576       | 61120478     | 22.7G | userdata |

- uboot Partition: For the uboot.img compiled from uboot.
- misc Partition: For misc.img, used by recovery.
- boot Partition: For the boot.img compiled from the kernel.
- recovery Partition: For the recovery.img compiled from recovery.
- backup Partition: Reserved, not in use at the moment.
- rootfs Partition: For the rootfs.img compiled from buildroot, debian, or yocto.
- oem Partition: For manufacturers to use, storing manufacturer's apps or data. Mounted in the /oem directory.
- userdata Partition: For apps to temporarily generate files or for end users, mounted in the /userdata directory.